A.D.A. Controller panel · general arrangement

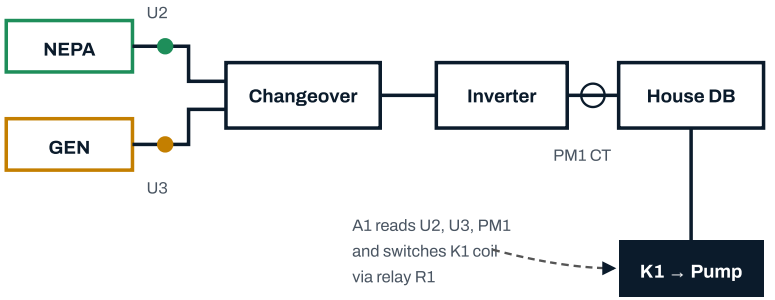
Concept layout for one building · Drawing ADA-PNL-01 rev A · November 2026
Not to scale · Final design and wiring by a qualified electrician



PARTS KEY

Q1	**Panel isolator** 2-pole. Switches the whole panel off for maintenance.
F1	**6 A MCB** protecting the controller supply.
PS1	**DIN power supply**, 230 V AC to 24 V DC for the controller.
PM1	**PZEM-004T meter**, CT clamp on the house live: voltage, current, energy.
U2/U3	**AC detection** on the NEPA and GEN inputs (before the changeover). Neither live but house powered = **inverter**.
A1	**DIN ESP32 controller** with 4 relays and inputs. Talks to the hub over Wi-Fi.
K1/K2	**Contactors**. A1's relay only switches the coil; the contactor carries the pump or AC current.
X1	**Terminals** for every incoming and outgoing wire, plus earth.

HOW IT CONNECTS



WIRE COLOURS

- Live (brown)
- Earth (green/yellow)
- DC 0 V
- Neutral (blue)
- 24 V / 5 V DC +
- Signal / control

Electrician only for 230 V. Q1, F1, PS1, PM1, U2, U3, K1/K2 and all mains terminals must be installed by a qualified electrician to local regulations. Size cables and contactors to the actual pump rating. If A1 loses contact with the hub, relays release and the pump stops (fail-safe).

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